SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Robotics and Automation

Project Based Learning

Batch 2023-27, Semester VI

Final Progress Presentation

EMG-Controlled 3D-Printed Prosthetic Hand with Myoelectric Actuators

Group ID: 3

Name of Students (PRN)

Priyanshu Mohanty	*23070127045*
Tej Datir	*23070127062*
Ayush Chhaparia	*23070127015*
Shounak Chaudhari	*23070127070*

Guide and Co-guide Name

Dr Satish Kumar
Professor, PhD In-charge
Department of Robotics and Automation
Symbiosis Institute of Technology, Pune

Dr Ramesh B.T
Assistant Professor
Department of Robotics and Automation
Symbiosis Institute of Technology, Pune

AY 2025-26

Introduction

Background / Rationale of the Study

Upper limb loss significantly affects independence and quality of life. Although advanced prosthetic technologies exist, their high cost and complexity make them inaccessible to a large population, especially in developing regions. This creates a critical need for affordable, functional, and user-friendly assistive solutions.

Significance of the Study

This project aims to bridge the accessibility gap by developing a low-cost, EMG-controlled, 3D-printed prosthetic hand. By leveraging readily available components and modular design, the system enhances functional independence while promoting inclusivity, affordability, and technological accessibility.

Current Trends

- Adoption of myoelectric (EMG) control for intuitive prosthetic operation
- Integration of embedded systems and smart sensors for real-time response
- Use of additive manufacturing (3D printing) for customization and cost reduction
- Movement toward open-source, modular prosthetic designs

Review of Literature

Sr. No.	Authors	Publication	Title of Paper	Key Findings
1	C Gentile	MDPI Review	A Perspective on Prosthetic Hands Control: From the Brain	Neuroscience perspective on prosthetic control interfaces and brain-driven strategies.
2	(Francisco) E. D. Papaleo et al.	PMC Article	Integration of Proprioception in Upper Limb Prostheses	Highlights importance of sensory (intraneural) feedback to improve grip control.
3	Taha Beyrouthy et al.	Advances in Sci., Tech & Eng. Systems J.	EEG Mind Controlled Smart Prosthetic Arm – A Comprehensive Study	Review of EEG-controlled prosthetic systems and signal processing.
4	Pranesh Sathish Kumar	ArXiv	BIONIX: A Wireless, Low-Cost Prosthetic Arm with Dual-Signal EEG and EMG Control	Presents low-cost EEG + EMG neuromuscular control system.
5	Roman Akinshin et al.	ArXiv	NeuroManip: Prosthetic Hand Manipulation via EMG & Eye Tracking	Context-aware controller using EMG + eye tracking for improved gesture recognition.
6	Mehrshad Zandigohar et al.	ArXiv	Multimodal Fusion of EMG and Vision for Human Grasp Intent Inference in Prosthetic Hand Control	Fusion of EMG + vision improves grasp classification accuracy.
7	Keyan Shen et al.	Academic Journal of Science and Technology	Comparative Study on Different Control Methods of Limb Prostheses	Reviews EMG, neuromuscular, and hybrid vision + EEG-EMG control methods.

Review of Literature

Sr. No.	Authors	Publication	Title of Paper	Key Findings
8	Linda Resnik et al.	Journal of NeuroEngineering and Rehabilitation	Evaluation of EMG Pattern Recognition for Upper Limb Prosthesis Control	Compares EMG pattern recognition with direct myoelectric control effectiveness.
9	Z Chen	PMC & Open Review	A Review of Myoelectric Control for Prosthetic Hand	Systematic review of intention recognition and control strategies for myoelectric hands.
10	Sahla Yoosuf Husain Ahmed	IJERT	Bionic Hand	Open-access paper on bionic prosthetic hand design and fabrication.
11	M. S. Aspalli et al.	Zenodo / IEEE	Prosthetic Arm Using EMG Sensor	Reviews EMG sensor-based prosthetic arm systems and functionalities.
12	Mustafa Can Kadilar et al.	IJA Journal	EMG-based Prosthetic Hand Design and Control Through DTW	Proposes DTW algorithm to improve grip control accuracy in EMG prostheses.
13	Jack Tchimino et al.	Scientific Reports	Application of EMG Feedback for Hand Prosthesis Control in High-Level Amputation	Shows EMG feedback enhances force control even in high-level amputations.
14	Agamemnon Krasoulis et al.	Journal of NeuroEngineering and Rehabilitation	Improved Prosthetic Hand Control with Concurrent Use of Myoelectric and Inertial Measurements	Combining myoelectric and inertial measurements increases control robustness.
15	Robin Arbaud et al.	ArXiv	Learning and Online Replication of Grasp Forces from EMG Signals for Prosthetic Finger Control	Demonstrates EMG-driven force estimation model enabling precise grip control.

Research Gap

Identified Gaps in Existing Prosthetic Systems:

- Lack of **affordable, fully integrated upper-limb prosthetic designs** suitable for large-scale accessibility
- Limited **open-source and reproducible solutions** for academic and research use
- Mechanical design and control systems are often developed **independently rather than as a unified system**
- Absence of **standardized performance evaluation metrics** for grip response, accuracy, and signal reliability
- Insufficient documentation on **integration of modern EMG sensors (e.g., MyoWare) with ESP32** for real-time control
- Most low-cost prototypes lack **modularity, scalability, and user adaptability**

Statement of the Problem

- There is a lack of low-cost, modular, and fully integrated upper-limb prosthetic systems that combine mechanical design, myoelectric control and performance validation within the practical and financial constraints of undergraduate engineering development.

- Existing commercial prosthetics are cost-prohibitive, while academic prototypes often lack full system integration, reproducibility, and standardized evaluation.

- Therefore, there is a need to design and validate an accessible EMG-controlled prosthetic hand system that is affordable, modular, reproducible, and functionally tested using measurable criteria.

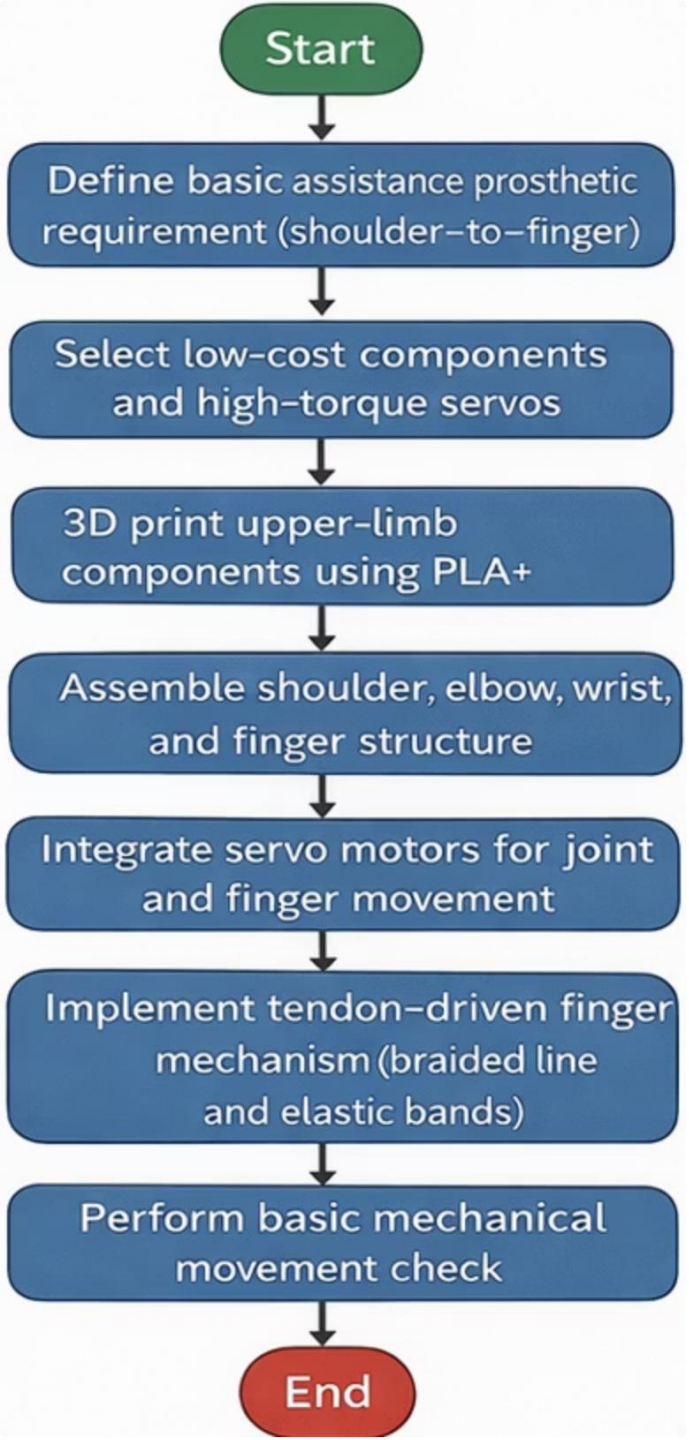
Objectives

- To develop a shoulder-to-finger 3D printed upper limb prosthetic using components and high-torque servos that fall under basic assistance design without clinical prosthetic development or fitting purposes.
- To implement a workable control system based on thresholds from EMG sensors and an Arduino-class microcontroller (ESP32) so that signals from muscle detection can appropriately control the prosthetic without additional complications.
- To evaluate the prosthetic on a physical model/rig according to functional parameters including but not limited to grip responsiveness, control accuracy, signal reliability, and user-friendliness.

Methodology

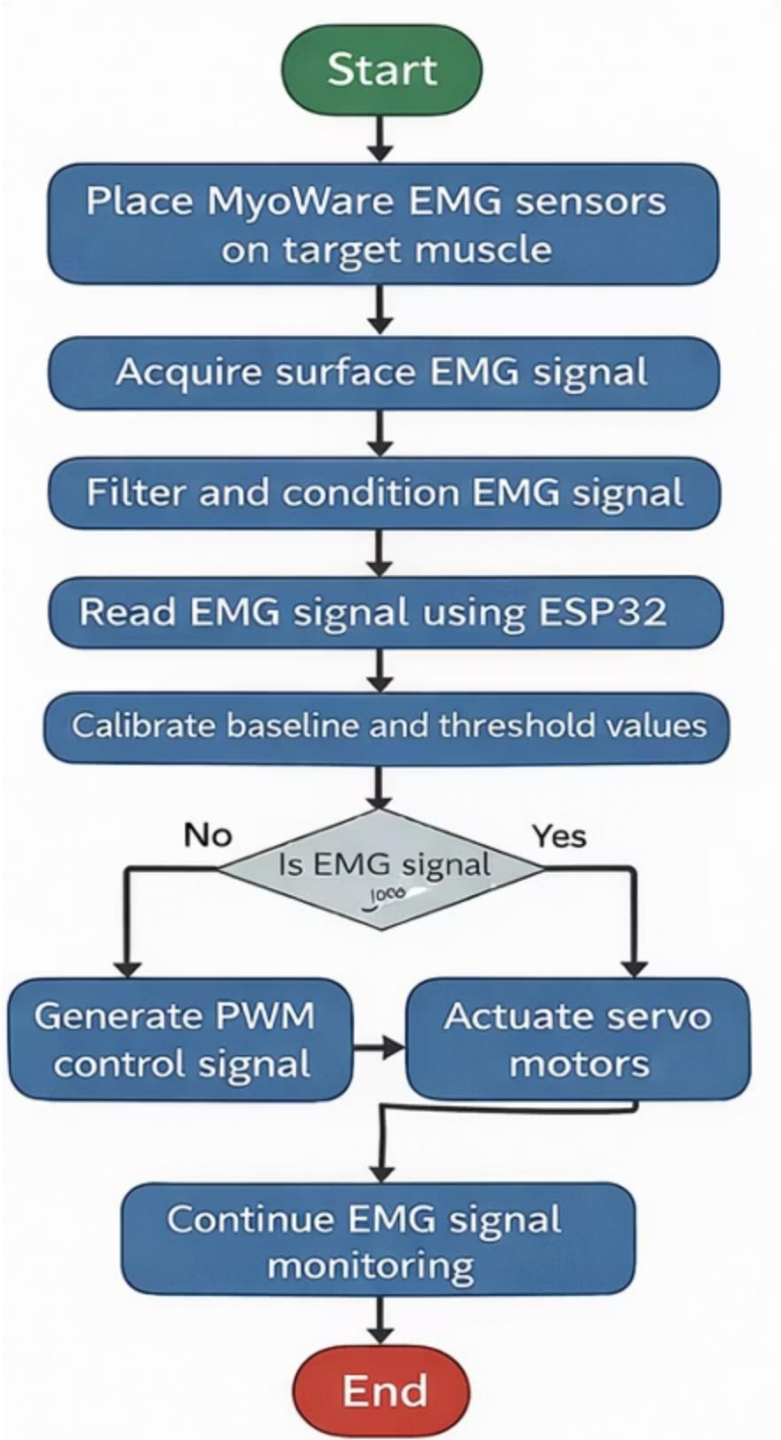
System Development Approach

- **Requirement Analysis & System Design**
  Define functional requirements and overall prosthetic architecture.
- **Mechanical Design & 3D Printing**
  Design and fabricate prosthetic components using PLA+ for lightweight and durability.
- **Electronics Integration**
  Integrate ESP32, EMG sensors, servo motors, and power regulation modules.
- **Firmware Development**
  Implement EMG signal acquisition, filtering, threshold detection, and servo control.
- **System Integration & Calibration**
  Assemble components and calibrate EMG thresholds for reliable control.
- **Testing & Performance Evaluation**
  Evaluate grip responsiveness, control accuracy, signal reliability, and usability.



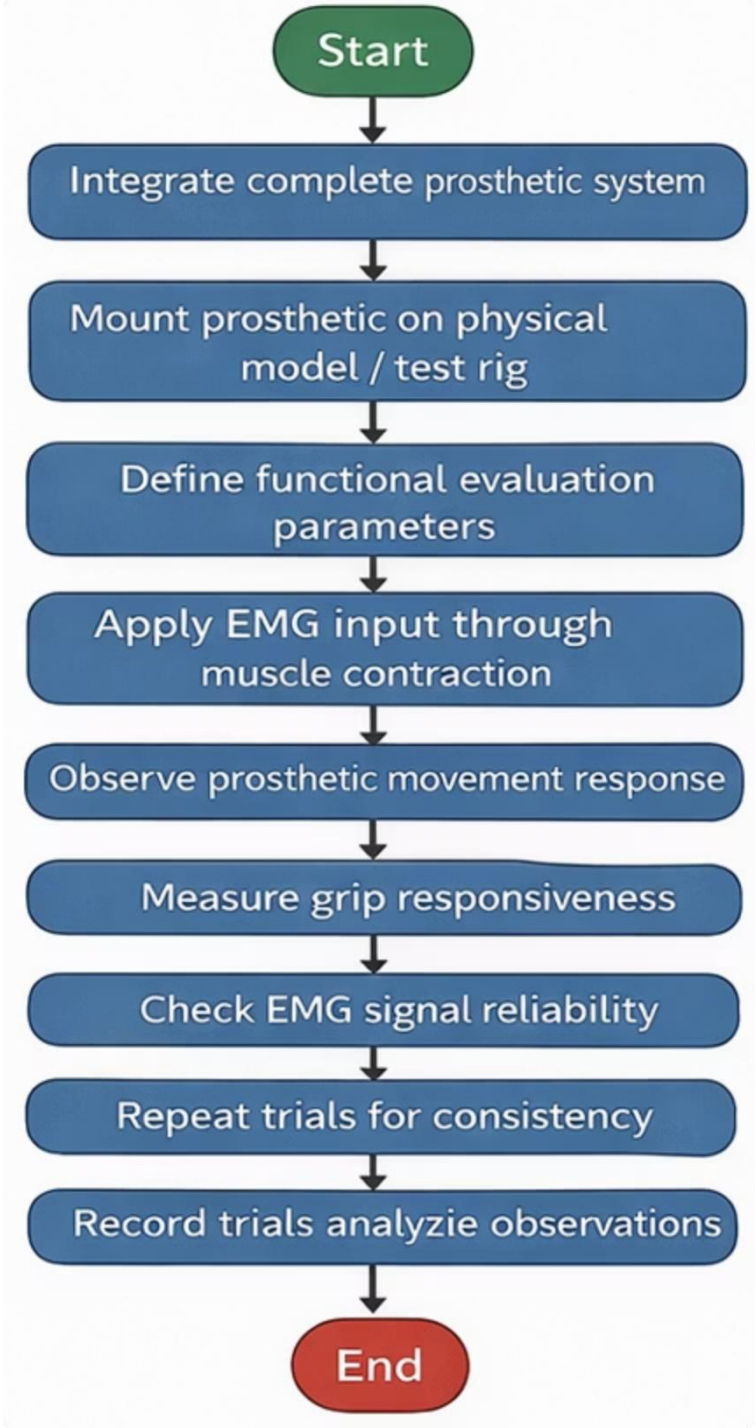
OBJECTIVE-1

To develop a shoulder-to-finger 3D-printed upper limb prosthetic using components and high-torque servos that fall under basic assistance design without clinical prosthetic development or fitting purposes.



OBJECTIVE-2

To implement a workable control system based on thresholds from MyoWare EMG sensors and an Arduino-class microcontroller (ESP32) so that signals from muscle detection can appropriately control the prosthetic without additional complications.



OBJECTIVE-3

To evaluate the prosthetic on a physical model/rig according to functional parameters including grip responsiveness, control accuracy, signal reliability, and user-friendliness.

Tools and Techniques

Hardware Tools

- **ESP32 Development Board** – Signal processing and control
- **EMG Sensors** – Muscle signal acquisition
- **MG996R & MG90S Servo Motors** – Joint and finger actuation
- **3D Printer (FDM)** – Fabrication of prosthetic components
- **3S Li-Po Battery & Buck Converter** – Portable and regulated power supply

Software Tools

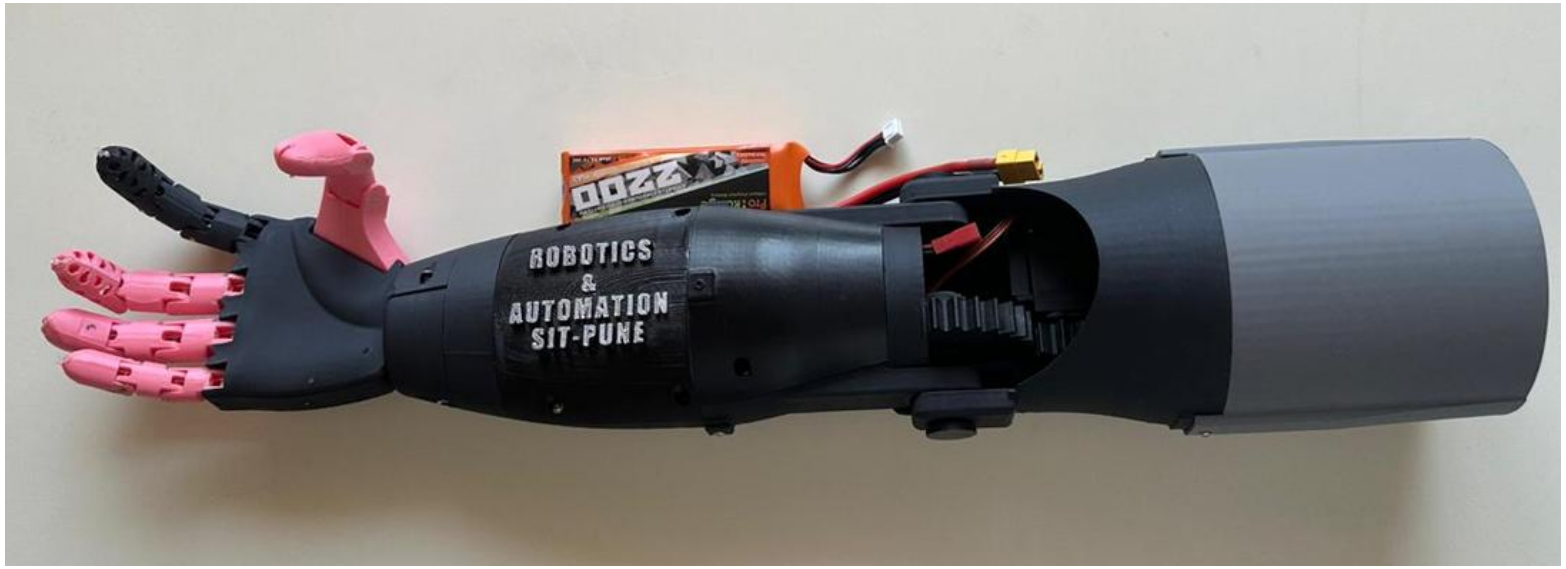
- **CAD Software (Fusion 360 / SolidWorks)** – Mechanical design
- **Arduino IDE** – ESP32 programming
- **Serial Plotter** – EMG signal visualization and calibration

Key Techniques Implemented

- **Surface EMG Signal Acquisition**
  Detect muscle activity using non-invasive electrodes.
- **Threshold-Based Myoelectric Control**
  Trigger prosthetic movement when EMG signals exceed preset thresholds.
- **PWM-Based Servo Actuation**
  Control servo position using pulse width modulation from ESP32.
- **Tendon-Driven Finger Mechanism**
  Use braided fishing line for flexion and elastic bands for extension.
- **Physical Rig-Based Evaluation**
  Test system performance using measurable functional parameters.

Results and Discussion

The prototype demonstrates successful fabrication and assembly of the 3D-printed prosthetic hand, including finger modules and forearm housing. The tendon-driven mechanism enabled smooth finger articulation, validating the mechanical design and servo integration. Initial tests confirmed proper alignment of components and sufficient range of motion for basic grasping functions. The modular structure allows easy maintenance and further integration of electronics and control systems, supporting the feasibility of a low-cost, functional prosthetic solution.



- Individually actuated fingers enable human-like gripping and motion
- Fingers are likely driven by servo motors for precise control
- 3D printed structure ensures lightweight and customizable design
- Easy assembly due to modular and compact construction

This prototype highlights:

- Integration of mechanical design and embedded systems
- Real-time control of actuators
- Compact and ergonomic design
- Application potential in prosthetics, automation, and robotic manipulation

Outcomes/Results:

Achieved Outcomes

- Successful design and fabrication of a **3D-printed prosthetic hand structure**
- Reliable acquisition of muscle signals using **MyoWare EMG sensors**
- Implementation of **ESP32-based control system** for real-time signal processing
- Accurate servo actuation enabling **finger movement and basic grasping functions**
- Integration of tendon-driven mechanism for **lightweight and natural motion**

System Performance Observations

- Responsive finger movement corresponding to muscle contractions
- Stable EMG signal detection with proper calibration
- Smooth servo operation with effective PWM control
- Modular design enabling easy maintenance and component replacement

Key Project Achievements

- Developed a **low-cost prosthetic prototype (~₹13,000)**
- Demonstrated feasibility of **affordable myoelectric prosthetic systems**
- Established a reproducible platform for **educational and research applications**

References

1. Gentile, C. *A Perspective on Prosthetic Hands Control: From the Brain.* MDPI Review.
2. Papaleo, F. E. D., et al. *Integration of Proprioception in Upper Limb Prostheses.* PMC Article.
3. Beyrouthy, T., Al Kork, S., Akl Korbane, J., et al. *EEG Mind Controlled Smart Prosthetic Arm – A Comprehensive Study.* Advances in Science, Technology & Engineering Systems Journal.
4. Sathish Kumar, P. *BIONIX: A Wireless, Low-Cost Prosthetic Arm with Dual-Signal EEG and EMG Control.* ArXiv.
5. Akinshin, R., Lopatina, E., Bogatikov, K., et al. *NeuroManip: Prosthetic Hand Manipulation via EMG & Eye Tracking.* ArXiv.
6. Zandigohar, M., Han, M., Sharif, M., et al. *Multimodal Fusion of EMG and Vision for Human Grasp Intent Inference in Prosthetic Hand Control.* ArXiv.
7. Shen, K., Zhang, Z., Guo, S. *Comparative Study on Different Control Methods of Limb Prostheses.* Academic Journal of Science and Technology.
8. Resnik, L., Huang, H., et al. *Evaluation of EMG Pattern Recognition for Upper Limb Prosthesis Control.* Journal of NeuroEngineering and Rehabilitation.
9. Chen, Z. *A Review of Myoelectric Control for Prosthetic Hand.* PMC & Open Review.
10. Ahmed, S. Y. H. *Bionic Hand.* IJERT.
11. Aspalli, M. S., Iqbal, M. O., et al. *Prosthetic Arm Using EMG Sensor.* Zenodo / IEEE.
12. Kadilar, M. C., Toptas, E., Akgun, G. *EMG-based Prosthetic Hand Design and Control Through DTW.* IJA Journal.
13. Tchimino, J., Hansen, R. L., et al. *Application of EMG Feedback for Hand Prosthesis Control in High-Level Amputation.* Scientific Reports.
14. Krasoulis, A., Suphi Erden, M., et al. *Improved Prosthetic Hand Control with Concurrent Use of Myoelectric and Inertial Measurements.* Journal of NeuroEngineering and Rehabilitation.
15. Arbaud, R., Motta, E., Avaro, M. D., et al. *Learning and Online Replication of Grasp Forces from EMG Signals for Prosthetic Finger Control.* ArXiv.